

Mustafa Alp Ekici

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EDUCATION

Politecnico di Milano - MSc in Electronics Engineering

Sept 2026 - Present

Milano, Italy

Middle East Technical University - BSc in Electrical and Electronics Engineering

Sept 2022 - June 2026

- GPA: 3.61/4.00
- Specialization: Electronics, Biomedical

Middle East Technical University - BSc in Computer Engineering

Sept 2020 - June 2022

- Transferred to Electrical and Electronics Engineering Department

EXPERIENCE

Research Intern, ERASMUS+ Traineeship - TU Delft, Delft, Netherlands

July 2025 - Sept 2025

- Designed and simulated Love-mode SAW biosensors in COMSOL; analyzed guiding-layer thickness through eigenfrequency studies.
- Developed MATLAB biofilm-formation algorithms and a COMSOL LiveLink workflow for automated 3D biofilm generation.
- Optimized sensor geometry and materials, correlating post-integration phase shifts with bacterial quantity.

Part-time Engineer - METU MEMS Center, Ankara, Turkey

Sept 2024 - April 2025

- Researched delta-sigma ADC architectures, readout circuitry for micro-g MEMS accelerometers, and sensor production methods.

Engineering Intern - Roketsan, Ankara, Turkey

Aug 2024 - Sept 2024

- Applied systems-engineering principles to avionics and designed DC-DC converters and Pi filters in LTspice.

Research Intern - UMRAM, Ankara, Turkey

July 2024

- Designed RF bias tees and low-noise amplifiers for 3T MRI systems using Altium Designer and Proteus; performed PCB design, soldering, and network-analyzer testing.

PROJECTS

Analog IC Design: Operational Amplifier, Bandgap Reference & LDO

- Designed and simulated a two-stage CMOS op-amp, bandgap reference, and LDO in XFAB 180 nm CMOS using Cadence.
- Verified over 97 dB DC gain, over 5 MHz unity-gain bandwidth, over 93 dB CMRR, approximately 1.794 V output, 1.4% bandgap variation, over 60 degrees phase margin, and up to -67 dB PSRR at 10 kHz.

Review on Silicon-Germanium (SiGe) Technology

- Reviewed SiGe HBT bandgap engineering, strain physics, fabrication methods, reliability, and sub-THz applications.

Neural-Network MAC Tile Accelerator Design (VLSI)

- Completed RTL-to-layout flow on XFAB 180 nm with Cadence Genus and Innovus; a dual-MAC architecture achieved 2x throughput and 44% energy reduction.

X-Ray CT Simulation & Image Reconstruction

- Developed MATLAB forward-projection and inverse-reconstruction algorithms; filtered backprojection reduced reconstruction error by around 80%.

Micro Air Conditioner Designing

- Designed and prototyped an analog temperature-control system in LTspice with less than plus/minus 0.8 degrees C regulation error.

Power Cable Selection Interface Design

- Built a Python GUI with PyQt5 and Pandas to automate cable selection, electrical calculations, correction factors, and a 10-year economic analysis.

SKILLS

Programming & Software: Sentaurus TCAD, Cadence, Python (PyQt5, Pandas), MATLAB, COMSOL Multiphysics (LiveLink), LTspice, Altium Designer, Proteus, KiCad

LANGUAGES

Turkish: Native **English:** C2 **German:** A2

CERTIFICATION

Cleanroom Training Certificate - METU MEMS Center

Trained and certified for cleanroom work, including entry procedures, safety rules, and proper dressing.